Luhuan Wu

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Academic Positions

Johns Hopkins University — Baltimore, MD
Department of Applied Mathematics and Statistics
Assistant Professor — 2026 - Current

Flatiron Institute — New York, NY
Center for Computational Mathematics
Associate Research Scientist — 2025 - 2026

Education

Columbia University — New York, NY
Ph.D. in Statistics. Advisors: John P. Cunningham and David M. Blei — 2020 - 2025
M.Phil. in Statistics — 2020 - 2024
M.S. in Data Science — 2018 - 2020

Nanjing University — Nanjing, China
B.S. in Mathematics — 2014 - 2018

Publications & Preprints

* for equal contribution

1. Extending Pathwise Gradients to Discrete Random Variables via Finite-Order Relaxation. *arXiv:2610.07786*
   Donghan He, **Luhuan Wu**
2. Bridging Stochastic Flow Maps and Boltzmann Generators with Normalizing Flows. *arXiv:2609.31978*
   Louis Grenioux, Ruikang OuYang, **Luhuan Wu**
3. Robust Inference-Time Steering of Protein Diffusion Models via Embedding Optimization. *NeurIPS 2026*
   Minhuan Li, Jiequn Han, Pilar Cossio, **Luhuan Wu**
4. Reverse Diffusion Sequential Monte Carlo Samplers. *NeurIPS 2025*
   Luhuan Wu, Han Yi, Christian A. Naesseth, John P. Cunningham
5. Bayesian Modeling of Data from Heterogeneous Environments. *arXiv:2506.22675*
   Luhuan Wu, Mingzhang Yin, Yixin Wang, John P. Cunningham, and David M. Blei
6. Posterior Uncertainty Quantification in Neural Networks using Data Augmentation. *AISTATS 2024*
   Luhuan Wu, Sinead Williamson
7. Practical and Asymptotically Exact Conditional Sampling in Diffusion Models. *NeurIPS 2023*
   Luhuan Wu*, Brian L. Trippe*, John P. Cunningham, and David M. Blei

8. Denoising Deep Generative Models. *NeurIPS 2022 I Can't Believe It's Not Better Workshop*
Gabriel Loaiza-Ganem, Brendan Leigh Ross, **Luhuan Wu**, John P. Cunningham, Jesse C. Cresswell, Anthony L. Caterini
9. Variational Nearest Neighbor Gaussian Processes. *ICML 2022*
Luhuan Wu, Geoff Pleiss, and John P. Cunningham
10. Bias-free Scalable Gaussian Processes via Randomized Truncations. *ICML 2021*
Andres Potapczynski*, **Luhuan Wu***, Dan Biderman*, Geoff Pleiss, and John P. Cunningham
11. Hierarchical Inducing Point Gaussian Process for Inter-domain Observations. *AISTATS 2021*
Luhuan Wu*, Andrew Miller*, Lauren Anderson, Geoff Pleiss, David M. Blei, and John P. Cunningham
12. Inverse Articulated-body Dynamics from Video via Variational Sequential Monte Carlo. *NeurIPS 2020 Workshop on Differentiable Vision, Graphics, and Physics in Machine Learning*
Dan Biderman, Christian A. Naesseth, **Luhuan Wu**, Taiga Abe, Alice C. Mosberger, Leslie J. Sibener, Rui Costa, James Murray, John P. Cunningham
13. Variational Objectives for Markovian Dynamics with Backward Simulation. *ECAI 2020*
Antonio Khalil Moretti*, Zizhao Wang*, **Luhuan Wu***, Iddo Drori, Itsik Pe'er

Professional experience

Apple, AIML — Seattle, WA
Research Intern — May 2023 - Sept 2023

Columbia University, Zuckerman Institute — New York, NY
Research Staff — Feb 2020 - Sept 2020

Academic Service

Reviewer for JMLR (2021,2023,2025), ICML (2023,2024), NeurIPS (2022,2023,2025,2026), ICLR (2024,2025,2026,2027), AISTATS (2022,2023,2024), UAI (2024,2025), AAAI (2025), TMLR (2025), Statistics and Computing (2025, 2026), Electronic Journal of Statistics (2026), the American Statistician (2026), Bayesian Analysis (2026)

Talks and Panel Discussions

Nov 2026: SIAM Conference on Imaging Science. *Salt Lake City, UT*

Oct 2026: Mathematical Institute for Data Science, Johns Hopkins University. *Baltimore, MD*

Oct 2026: Department of Electrical and Computer Engineering, Johns Hopkins University. *Baltimore, MD*

Oct 2026: Department of Applied Mathematics and Statistics, Johns Hopkins University. *Baltimore, MD*

Sept 2026: Bay Area Light Sources Joint Users' Meeting, SLAC National Accelerator Laboratory. *Menlo Park, CA*

Aug 2026: Microsoft Research New England Generative Modeling & Sampling Workshop. *Cambridge, MA*

Mar 2026: Monte Carlo Online Seminar. *Virtual*

Dec 2025: UK Probabilistic AI "Prob_AI" Online Seminar. *Virtual*

Dec 2025: NeurIPS 2025 Workshop on Structured Probabilistic Inference & Generative Modeling. *San Diego, CA*

Nov 2025: Department of Statistics, University of Warwick. *Coventry, UK*

Oct 2025: Department of Statistics, Rutgers University. *New Brunswick, NJ*

Aug 2025: Microsoft Research New England Generative Modeling and Sampling Seminar. *Cambridge, MA*

Jun 2025: The Gatsby Tri-center Annual Meeting. *London, UK*

Apr 2025: The 7th Advances in Approximate Bayesian Inference. *Singapore*

Jun 2024: The 33rd ICSA Applied Statistics Symposium. *Nashville, TN*

Apr 2023: Berkeley–Columbia Meeting in Engineering and Statistics. *New York, NY*

May 2022: The 35th New England Statistical Symposium. *Storrs, CT*